Maui – Hurricane ESF-6 playbook

Data as of 2026-10-28

County: Maui · **Hazard:** Hurricane

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

People at risk **164,815**	Shelter demand (central) **4,944**	Shelter demand low – high **3,296 – 8,241**
Limited English (C) **4,067**	Power-dependent (M) **1,074**	Disability (I) **17,024**
Age 65+ (I) **31,848**	People without a vehicle (T) **7,284**	Known 2026 sites usable **7**

Shelter-seeking rates: low 2%, central 3%, high 5.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

2. Scope: county total vs. one community

County total = upper bound. It assumes every at-risk tract is hit at once (all tracts). Real events usually hit fewer communities.

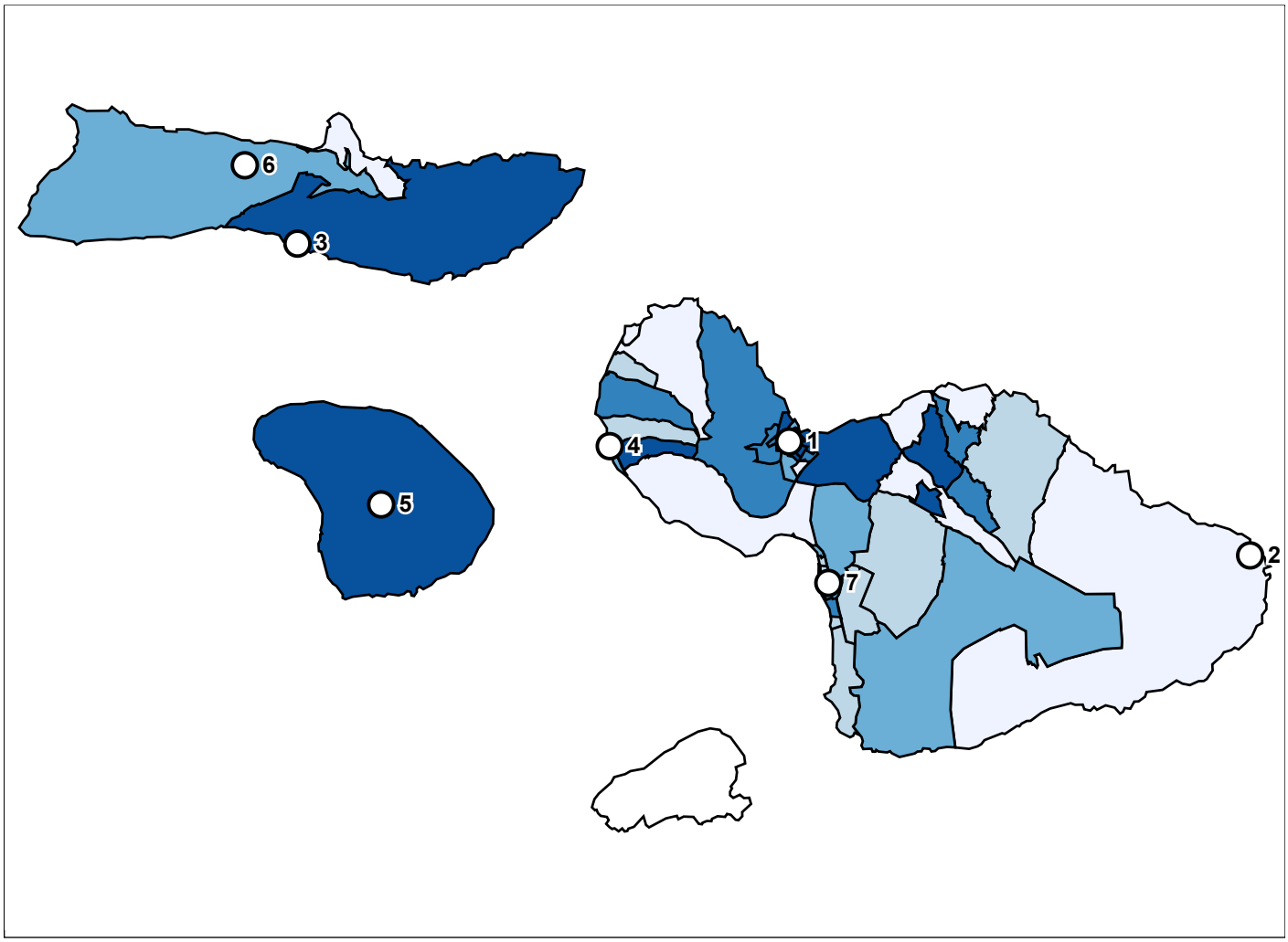
Largest single community (highest-demand census tracts)

Hurricanes affect the whole county, so these are simply the most populous at-risk tracts.

Tract	At risk	Shelter low / central / high	Limited English	Power-dep.	Disability	65+	No vehicle
Tract 308 · near Waihee-Waiehu	8,062	161 / **242** / 403	6	37	669	1,100	95
Tract 309.03 · Waihee-Waiehu	7,636	153 / **229** / 382	264	50	849	1,476	385
Tract 310.01 · Wailuku	6,977	140 / **209** / 349	87	30	622	897	192

Plan a realistic one-community event with the first row.

3. Map: AFN need and known sites



- AFN need 0–258 people (class 1 of 5)
- AFN need 258–392 people (class 2 of 5)
- AFN need 392–647 people (class 3 of 5)
- AFN need 647–1,016 people (class 4 of 5)
- AFN need 1,016–1,924 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

4. Known sites (opened in 2026 events)

#	Status	Site	Address	Reason	Times opened 2026
1	✓ **Usable**	Baldwin High School	1650 Kaahumanu Ave., Wailuku	no site-level screen for this hazard	1
2	✓ **Usable**	Hāna High & Elementary School Cafeteria	4111 Hāna Highway, Hāna	no site-level screen for this hazard	1
3	✓ **Usable**	Kilohana Recreation Center	344 Kamehameha V Highway, Kaunakakai	no site-level screen for this hazard	1
4	✓ **Usable**	Lahaina Civic Center Gym	1840 Honoapiʻilani Highway, Lahaina	no site-level screen for this hazard	2
5	✓ **Usable**	Lānaʻi Youth Center	101-165 Fifth St., Lānaʻi City	no site-level screen for this hazard	1
6	✓ **Usable**	Molokaʻi Baptist Church	495 Puupeelua Ave., Hoolehua	no site-level screen for this hazard	1
7	✓ **Usable**	South Maui Community Park Gym	1501 Liloa Drive, Kīhei	no site-level screen for this hazard	2
8	**Not located**	Mayor Hannibal Tavares Community Center (Social Hall)	91 Pukalani St., Pukalani	not located – not on map	1

Candidate public schools in this county usable for this hazard: **34**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

Need	Quantity	Unit	Proposed lead	Proposed support	Pledged	Gap
Cots / sleeping space	4,945	cots	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤72 h)	98,889	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤2 weeks)	197,778	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Water	4,945	gal/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Meals	9,889	meals/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Toilets (≤72 h)	124	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Toilets (≤2 weeks)	248	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Shelter staff	334	staff per shift	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community	—	confirm with partner
Power outlets for power-dependent residents	33	outlets	doh	county_em; health_care; utilities	—	confirm with partner
Accessible transport	7,284	people in no-vehicle households at risk	county_em	dhs; hiema; dot; business_community; other_agencies	—	confirm with partner
Pet space – dog households	733	households with a dog	county_em	hawaii_voad	—	confirm with partner
Pet space – cat households	428	households with a cat	county_em	hawaii_voad	—	confirm with partner
Children's area	1,058	children in shelter	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy	—	confirm with partner
Interpreters – Ilocano	51	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	28	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Spanish	9	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Japanese	7	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Marshallese	4	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Hawaiian	3	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Central Pacific hurricane season start (June 1) NOAA Central Pacific Hurricane Center	Confirm partner capacity pledges (cots, meals, generators, accessible vehicles, interpreters) and the list of sites usable for wind; refresh priority-outreach tracts.	County Emergency Management (each county)	dhs, hiema, red_cross
BEFORE	Hurricane Watch NOAA NWS / Central Pacific Hurricane Center	Contact priority tracts (power-dependent, no-vehicle, limited English); confirm which shelters will open, generator power for medical devices, and accessible transport.	County Emergency Management (each county)	doh, utilities, dot, dhs
BEFORE	Hurricane Warning NOAA NWS / Central Pacific Hurricane Center	Open shelters including AFN-capable sites; start accessible transport for residents without vehicles; send pre-approved multilingual messages.	County Emergency Management (each county)	red_cross, dhs, doe, dot
AFTER	Warning cancelled or all clear County / NWS / PTWC	Feeding, re-power for medical devices, recovery sheltering for displaced households; coordinate donations and volunteers.	American Red Cross	hawaii_voad, salvation_army, faith_based, utilities

7. Priority outreach and languages

Top 10 census tracts by AFN need (tract-level counts only)

#	Tract	At risk	Power-dep.	Disability	Limited English	No vehicle	AFN need
1	Tract 311.02 · Kahului	5,916	58	803	393	670	1,924
2	Tract 314.05 · Lahaina	6,442	33	599	438	812	1,882
3	Tract 319 · Kahului	5,325	29	609	489	517	1,644
4	Tract 309.03 · Waihee-Waiehu	7,636	50	849	264	385	1,547
5	Tract 311.06 · Kahului	4,690	29	547	259	358	1,193
6	Tract 317 · rural Molokaʻi	4,827	35	820	19	237	1,111
7	Tract 304.04 · Haliimaile	6,005	38	818	120	109	1,085
8	Tract 316.01 · Lanai City	3,218	22	513	175	368	1,079
9	Tract 304.06 · Pukalani	6,552	48	683	112	219	1,062
10	Tract 311.05 · Kahului	6,530	53	542	331	90	1,016

Languages: people who speak English less than "very well" (Kauai + Maui)

Kauaʻi and Maui share one language list: ACS PUMS reports both counties (and Kalawao) as a single area, PUMA 00100, so they cannot be separated.

Language	People 5+	90% MOE
Ilocano	9,131	±1,439
Tagalog, Filipino	5,074	±926
Spanish	1,711	±554
Japanese	1,339	±417
Marshallese	633	±488
Hawaiian	515	±289

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6

Source	Fetched	URL / file
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
Census TIGERweb 2020 Census Designated Places (tract community labels)	2026-10-27	https://tigerweb.geo.census.gov/arcgis/rest/services/TIGERweb/tigerWMS_Census2020/MapServer/28
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.